

Home-Brew PCBs

I am doing a project on the general-purpose printed circuit board (PCB) but not satisfied with the result. It's tough and the circuit doesn't work. Can you please guide me on how to make a PCB and use software to draw the layouts?

Hemal Chevli
Surat, Gujarat

EFY: You may refer to the 'Making Home-Brew PCBs' article published in EFY's April 2006 issue. PCB layout design software are included in the EFY-CD of almost every issue. You may try out such software as Eagle, DIP Trace and Express PCB. To purchase these EFY issues and CDs, dial EFY associates Kits'n'Spares at 011-26371661.

EFY-CD Software

The EFY-CD contains an older version of Acrobat Reader. Please make Acrobat Reader Version 9 available to EFY subscribers.

Kamal Jeet Singh
Hoshiarpur, Punjab

EFY: Thanks for the suggestion. We will try to include Version 9 at the earliest possible, permission for which is being sought.

Ultrasonic Travel Aid

I am developing an ultrasonic travel aid for the blind, for which I need the 'Microcontroller-based Ultrasonic Distance Meter' project published in EFY Feb. 2008. Could you please send me the project as there is no authorised distributor of EFY here in Nepal?

Sugat Ratna Tuladhar
Nepal

EFY: You can download the article from EFY website at 'www.efymagonline.com', by subscribing to Feb. 2008 issue using your credit card. Please feel free to contact us for any other information.

Rank Display System

In the 'Rank Display System for Race and Quiz Competitions' project published in EFY June 2008, I am facing the following problem: When power is switched on and switches S1 through S8 are pressed, it doesn't work. When we press any switch, it does not beep but switch S1 is stored permanently

(player 'A' is always first). Even if we press other switches, it does not change. The rank display is so fast that we can't identify the rank it displays. The problem is in the delay. I want to know the working of this project in detail. Please do let me know if there is any change required in the program.

Jayendra R.
Udupi, Karnataka

The author Pankaj Verma replies:

When power is switched on, the LCD displays as mentioned in the article. In order to debug your problem, just go through the flow-chart of the program. In your case, the program is perfectly doing the initialisation part. Then it should wait for any key to be pressed (see 'chk' subroutine). As soon as any key is pressed, the logic at port-0 will change from 'FFH' and the same logic will be moved to the accumulator. As a result, execution will jump to 'first' and the buzzer will sound.

If 'flag' is not set (timeout doesn't occur), chk1 subroutine will compare the accumulator with the value of the latch (port 2). Note that the accumulator is complemented before comparison (XORing, since ULN2803 inverts the port-0 output which is being fed back). After that, 'scan' subroutine checks for the bit which has been changed through pressing of the key. 'r0' is also incremented, which points to the position of the switch ('31H' through '38H' RAM locations) for the storage of the rank. 'r4' keeps on counting the number of iteration and hence the ranks (see 'next rank'). 'r5', 'r6' and 'r7' have been used to provide nested looping of timer-0 to produce exact 30 seconds of delay for timeout and three to four seconds for the buzzer.

Coming to your problem, the buzzer doesn't beep and always player A is first. If connection is not proper, only the buzzer will beep. Player A will always be the first if switch S1 is shorted permanently, or ULN2803 is faulty. Check it out. The flow of the program is perfect and the prototype works perfectly.

Emergency Light

The 'Fully Automatic Emergency

Light' circuit published in EFY Feb. 2007 uses a ferrite-core transformer. But generally most transformers come with an iron core. Can you please tell me why a ferrite core was used and what will happen if any other core material is used?

Prashanth Kumar
Hyderabad

The author Dr C.H. Vithalani replies:

Eddy current losses in transformers are proportional to the square of the thickness of the core strip. Therefore as operational frequency increases, we have to reduce the thickness of the core strip.

In the emergency light, the frequency used is 15 kHz, at which it is not practical to make thin laminations. Hence, instead of an iron core, a ferrite core is used to reduce eddy current losses.

Ferrite is a ceramic material having magnetic properties similar to silicon steel but a high receptivity. Hysteresis of ferrite is very small compared to iron.

Water Level Controller cum Motor Protector

The 'Water Level Controller cum Motor Protector' project published in EFY Feb. 2007 is not working properly. When the tank is empty, the motor operates normally, i.e., it starts after 90 seconds at constant voltage. But during voltage fluctuations (high/low voltage), when water is filled, it displays 'standby mode' and the motor doesn't run.

Firoz Ahmad
Through e-mail

The author Gursharanjeet Kalra replies:

When voltage is high or low, it goes into scanning mode and checks for further fluctuations. When the voltage stabilizes for at least 90 seconds, it jumps to standby mode. However, if you want it to go back from where it comes after receiving high or low signal, instead of using 'jump' (JNB, i.e. jump on no bit) command, you have to use 'call' command. Then all the addresses have to be recalculated.